Response to Reviewers

Paper No. S-28-4-002

Original title: Camerala: A Privacy-Preserving Browser-Native Framework for In-the-Wild Physiological Sensing and Task Measurement

We thank the handling editor and the reviewers for their careful reading and constructive comments. The manuscript was revised according to Reviewer 1's required Conditions 1–13. The revision narrows unsupported claims while preserving the intended contribution: a local-first browser-native framework for remote psychological/psychophysiological task sensing. We also thank Reviewer 2 for the positive evaluation.

Condition 1: The title of the manuscript should be reconsidered

Response. Thank you for this comment. The manuscript was revised to avoid implying that Camerala proposes a dedicated privacy-preserving algorithm. Following the overall revision, the title was changed to a more conservative local-first system title for remote psychophysiological task sensing.

Revision. The title no longer uses "Privacy-Preserving" or an unqualified "In-the-Wild" expression. Privacy is now discussed as an architectural privacy advantage of avoiding raw-video transmission, not as a dedicated privacy-preserving method. The new title also preserves the intended remote psychophysiological task-sensing use case.

Key revised text. "Camerala: A Local-First Browser-Native Framework for Remote Psychophysiological Task Sensing"

Location. Title page; Introduction; Contributions.

Condition 2: The necessity of real-time processing is not sufficiently explained

Response. Thank you for this important clarification. We revised the manuscript to explain that real-time processing is not claimed as the only technically possible architecture. We acknowledge that local video recording followed by offline processing could preserve timestamps if implemented carefully.

Revision. The revised manuscript clarifies that Camerala targets browser-based remote psychological and psychophysiological task experiments conducted in laboratory and home settings. In such settings, retaining identifiable facial video for later analysis can itself be a privacy and acceptability burden. Camerala therefore extracts face-image-derived records during task execution and stores only task events, exported feature fields, and auxiliary context fields, without raw facial video persistence. The role of real-time browser-side processing is thus feature-only local logging and timestamp alignment within the same client-side runtime, not online task adaptation or a claim that physiological estimates are more accurate.

Key revised text. "Local video recording followed by offline processing could preserve timestamps if implemented carefully. However, the intended use case of Camerala is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings, where retaining identifiable facial video for later analysis can itself be a privacy and acceptability burden."

Location. Introduction; System Architecture; Data Management and Local-First Persistence.

Condition 3: The Research Questions are not defined clearly enough

Response. Thank you for this comment. The Research Questions were rewritten to define the tested configuration and the observation criteria more concretely.

Revision. We revised RQ1 to ask whether Camerala's task-synchronized face-derived records, including rPPG-oriented ROI values, satisfy the operational criteria used for descriptive analysis in the intended remote psychological/psychophysiological task use case. The revised manuscript defines concrete operational criteria at the trial and exported-window levels: reaction times within 100–1500 ms for trial-level retention and `mean_quality` ≥ 0.8 for exported-window-level landmark availability. The completed deployment consisted of 600 trials in total. After applying the RT criterion, 541/600 trials were retained for descriptive analysis. All 299 exported windows had `mean_quality` $= 1.0$ and therefore satisfied the operational landmark-availability criterion.

Key revised text. "For RQ1, we evaluated the exported records using operational criteria at the trial and exported-window levels. At the trial level, trials were retained for descriptive analysis when reaction times fell within 100–1500 ms. At the exported-window level, landmark availability was evaluated using `mean_quality`, the mean of the binary landmark-availability flag within each 10-s window; exported windows with `mean_quality` ≥ 0.8 were treated as satisfying the operational landmark-availability criterion."

Location. Introduction, Research Questions.

Condition 4: It is unclear what the "between-environment differences" in RQ2 refer to

Response. Thank you for this important clarification. We retained the original intent of RQ2, which asks what kinds of between-environment differences are observable in the physiological signals collected through the deployment. However, the revised manuscript clarifies that the term "physiological signals" is used operationally to refer to task-synchronized face-derived and sensor-derived exported records, not validated physiological-state measurements. The Results and Discussion now interpret laboratory/home differences as deployment-context differences that may reflect environmental conditions, screen light, posture, camera geometry, task timing, the measurement pipeline, participant state, or their mixture, and not as psychological or physiological effects.

Revision. The Abstract, the explanatory text immediately following RQ2, Results, Discussion, and Conclusion were revised to state that the observed laboratory/home differences are deployment-context differences in exported records. Because the present design did not include reference physiological sensors, multiple participants, or controlled separation of lighting, screen light, posture, camera geometry, task timing, measurement-pipeline factors, and participant state, the observed differences are not interpreted as psychological, cognitive, motivational, or physiological effects. Future work requirements were added, including reference ECG or contact PPG, multiple participants, multiple devices, controlled lighting/camera/posture conditions, and manipulation checks.

Key revised text. "In RQ2, the term 'physiological signals' is used operationally to refer to task-synchronized face-derived and sensor-derived exported records, including rPPG-oriented ROI values and related exported fields. These records are not treated as validated physiological-state measurements."

"The present design was not able to determine whether laboratory/home differences reflected psychological state, physiological state, lighting, screen light, posture, camera geometry, the measurement pipeline, task timing, participant state, or their mixture. Therefore, laboratory/home differences were interpreted as deployment-context differences rather than psychological, cognitive, motivational, or physiological effects."

Location. Abstract; Introduction, Research Questions; Results; Discussion; Conclusion.

Condition 5: The novelty and positioning of the study are insufficiently justified

Response. Thank you for identifying the relevant prior work. We agree that browser-based, web-deployable, edge-based, and privacy-oriented rPPG systems have already been explored. The Related Work section was expanded, and the contribution claim was repositioned accordingly.

Revision. The Related Work section now includes all studies, systems, and public demonstrations explicitly identified by the reviewer: Gupta and Etemad, Ayesha et al., LabVanced rPPG, FacePhys-Demo, RhythmEdge, and Di Lernia et al. LabVanced and FacePhys-Demo are described as public tools/demonstrations rather than peer-reviewed empirical validation studies. We removed or substantially weakened contribution claims based on browser-based rPPG functionality, local rPPG processing, edge-oriented heart-rate estimation, or privacy preservation as an algorithm. The revised manuscript instead positions Camerala as a system-integration contribution for remote psychological/psychophysiological task experiments. Specifically, Camerala integrates browser-based task execution, face-derived feature extraction, timestamp alignment between task events and exported frame/window records, operational landmark-availability checking, and local CSV/ZIP persistence without raw facial video retention within one browser runtime. This work is presented as a limited system-development contribution and implementation example, not as a new rPPG estimator or a validated physiological measurement system.

Key revised text. "The contribution of Camerala is not browser-based rPPG functionality, local rPPG processing, edge-based heart-rate estimation, or privacy preservation as an algorithm. Instead, Camerala contributes a local-first browser-native implementation example that integrates remote task execution, face-derived feature extraction, task-event/frame-record timestamp alignment, operational landmark-availability checking, and local CSV/ZIP persistence without raw facial video retention for remote psychological/psychophysiological task experiments."

Location. Related Work; Contributions; System Architecture; References.

Condition 6: Unclear claim of robustness against rigid head motion

Response. Thank you for this comment. We removed unsupported claims of computational efficiency and robustness against rigid head motion. The revised manuscript presents Camerala as a deployment and logging architecture, not as a method for improving rPPG signal quality under motion.

Revision. The revised manuscript no longer claims that Camerala corrects motion-induced rPPG degradation or provides motion-robust rPPG estimation. Motion estimates are retained only as task-synchronized face-derived records for descriptive inspection. The Results and Discussion now treat motion estimates as deployment-context records rather than evidence of motion handling or rPPG signal correction.

Key revised text. "This feature is retained as a motion-related exported field for descriptive inspection of task-synchronized records. It is not an algorithmic correction for motion artifacts in rPPG and is not used to compensate for motion-induced rPPG degradation."

"Motion estimates are reported as exported face-derived records for descriptive comparison between the laboratory and home contexts. They are not interpreted as evidence of motion handling or correction of rPPG degradation."

Location. Related Work; System Architecture, Feature Extraction Logic; Results; Discussion; Limitations.

Condition 7: Insufficient description of the concrete usage scenario

Response. Thank you for this clarification request. We revised the manuscript to define Camerala's target use case as a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings, rather than continuous physiological sensing during ordinary daily activities.

Revision. The revised manuscript now states that the participant sits in front of a laptop, views task stimuli, and responds by key press. It also clarifies that Camerala is intended to support task-synchronized acquisition of face-derived records while reducing the need to transmit or retain privacy-sensitive raw facial video. The Methodology further states that the deployment was a browser-based task experiment conducted in one laboratory setting and one home setting, not unconstrained daily-life physiological sensing.

Key revised text. "The intended use case of Camerala is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings. In this use case, the participant sits in front of a laptop, views task stimuli, and responds by key press. Camerala is not intended for continuous physiological sensing during ordinary daily activities."

"The deployment was a browser-based task experiment conducted in one laboratory setting and one home setting. The participant sat in front of a laptop, viewed Gabor stimuli, and responded using the keyboard."

Location. Introduction; Methodology, Target Usage Scenario and Exploratory Deployment.

Condition 8: Definition of "in-the-wild" and "naturalistic physiological sensing" are not clear

Response. Thank you for this point. The manuscript was revised to avoid giving the impression that Camerala was evaluated for unconstrained daily-life or continuous physiological sensing.

Revision. Unqualified "in-the-wild" and "naturalistic" expressions were removed from the manuscript. The revised manuscript describes the study as a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings. We also clarified that the participant sat in front of a laptop, viewed Gabor stimuli, and responded by key press. The deployment was not continuous physiological sensing during ordinary daily activities.

Key revised text. "The intended use case of Camerala is a browser-based remote psychological/psychophysiological task experiment conducted in laboratory and home settings. In this use case, the participant sits in front of a laptop, views task stimuli, and responds by key press. Camerala is not intended for continuous physiological sensing during ordinary daily activities."

Location. Abstract; Introduction; Methodology; Discussion; Conclusion.

Condition 9: The description of the system hardware is insufficient for reproducibility

Response. Thank you for this comment. We added a hardware and software configuration table to improve reproducibility.

Revision. The revised manuscript reports the author-confirmed PC model, CPU, GPU, RAM, storage, operating system, display, browser names, built-in webcam, requested camera constraints, and camera-control behavior. We also explicitly mark deployment-time details that were not logged, such as exact browser versions when unavailable, the actual negotiated camera resolution/frame rate, and the exact auto-exposure and auto-white-balance states.

Key revised text. "Table 2 summarizes the hardware and software information available for the deployment. Values that were not logged during the deployment are explicitly marked as not logged rather than inferred from the revision-time environment."

Location. Methodology, Hardware and Software Configuration; Table 2.

Condition 10: The description of the experimental environments is insufficient

Response. Thank you for the detailed request. An environment and lighting conditions table was added.

Revision. We limited the environmental description to setup/check information and camera-derived records. The brightness range is not reported as lux or externally measured illuminance. The revised section reports the laboratory/research room and private home room, daytime sessions, windows present, curtains closed, general artificial indoor lighting, fixed screen brightness, and camera-derived brightness ranges observed during setup/checks.

Key revised text. "Because the deployment used camera-derived setup checks rather than external light-meter measurements, this paper reports brightness ranges only as author-confirmed setup/check information."

Location. Methodology, Deployment Environment and Lighting Conditions; Table 3.

Condition 11: The validity of the cognitive framing manipulation is questionable

Response. Thank you for this important clarification. We agree that the original wording could imply that the negative and positive instruction blocks successfully induced evaluation pressure or reward motivation, which cannot be supported by the present N=1 author-participant deployment.

Revision. The revised manuscript no longer treats the negative, neutral, and positive blocks as validated cognitive or motivational manipulations. They are repositioned as exploratory instructional task contexts used to demonstrate that Camerala can present different task-interface contexts while recording synchronized behavioral and face-derived records. The manuscript explicitly states that these contexts are not interpreted as inducing evaluation pressure, reward motivation, cognitive effects, motivational effects, or psychological effects.

Key revised text. "The task included negative, neutral, and positive instruction contexts as exploratory task-interface conditions. These contexts were used to demonstrate that Camerala can present different instruction frames while recording synchronized behavioral and face-derived records. The labels 'negative' and 'positive' refer only to instruction labels used in the task interface. Because the participant was the author and knew that the instructions were experimentally constructed, these contexts are not treated as validated manipulations of evaluation pressure, reward motivation, cognitive state, or motivational state."

"The instruction contexts were used only as task-interface contexts in the deployment. We do not report them as evidence that evaluation pressure, reward motivation, cognitive effects, or motivational effects were induced."

Location. Participant section; Experimental Procedure and Instructional Task Contexts; Results; Discussion and Limitations.

Condition 12: The central claim in the Discussion should be reconsidered

Response. Thank you for this comment. The Discussion was rewritten around what the system demonstrated under the tested deployment and what remains unresolved.

Revision. The revised Discussion focuses on the fact that the system was built and ran under laboratory and home settings, while the observed differences remain deployment-context differences that cannot isolate psychological or physiological effects. Future validation requirements and design safeguards were added.

Key revised text. "The results do not establish general deployability or physiological/cognitive effects. Instead, they motivate design requirements and safeguards for future local-first deployments."

Location. Discussion.

Condition 13: Several claims are not sufficiently supported and should be weakened

Response. Thank you for this guidance. The Abstract, Results, Discussion, Limitations, and Conclusion were revised to substantially weaken unsupported claims.

Revision. We rechecked the logs and analysis description and limited the revised results to indicators directly recomputable from the exported CSV files. The manuscript no longer states that the study "establishes" a design pattern. We replaced the ambiguous confidence-threshold wording with an implementation-faithful description of `mean_quality` as an exported-window-level landmark-availability ratio. The `quality` field is defined as a binary landmark-availability flag, and `mean_quality` is the average of this flag within each exported 10-s window. The current runtime forwards results to the logging callback only when FaceMesh landmarks are returned; therefore, the saved frame records in this deployment represent landmark-returned exported frames. Thus, `mean_quality` ≥ 0.8 means that landmarks were available in at least 80% of exported frame records in that window. This value is not a continuous MediaPipe confidence score, an indicator of rPPG measurement quality, or evidence of physiological measurement accuracy. In the present deployment, all 299 exported windows had `mean_quality` $= 1.0$; therefore, no window was excluded by this criterion. The RT-based retained-trial count was updated to 541/600 after applying the 100–1500 ms reaction-time criterion. The revised manuscript states that 600 trials were completed in total and that 541/600 trials were retained after the reaction-time criterion. The previous exposure-related term was replaced with a frame-to-frame green-channel brightness-change indicator. The manuscript also clarifies that this indicator is not a measure of camera exposure settings, external illuminance, environmental events, psychological or physiological state, or rPPG accuracy.

Key revised text. "For each exported 10-s window, `mean_quality` was computed as the mean of the `quality` flag over the exported frame records in that window. Thus, `mean_quality` describes landmark availability within exported frame/window records and not face-detection availability across all camera frames."

Location. Abstract; Results; Discussion; Limitations; Conclusion.

Reviewer 1 Other Comments

Response. Thank you for the additional comments on figure and table readability.

Revision. Table 1 was reformatted with clearer column widths. The previous longitudinal and task-result figures contained overstrong embedded labels, so they were removed from the revised manuscript rather than retained. The remaining results presentation relies on conservative text and tables without p-values, significance marks, or causal wording.

Location. Table 1; Results; removed previous Figure 2, Figure 3, and Figure 4 from the revised manuscript.